

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 899706156 | hariramakrrish@gmail.com | [LinkedIn](#) | Stamp 1G — Eligible to work full-time in Ireland

PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India **09/2021 – 01/2025 (3.5 Years)**

- Prepared and optimized **3D print files** using proprietary software, successfully producing over 500 prototype parts with a **98.5% first-pass yield** for customer demonstrations.
- Executed **3D scanning demonstrations** for potential clients, showcasing handheld and desktop scanner capabilities and gathering technical requirements for reverse engineering workflows.
- Performed intermediate **reverse engineering** tasks, including mesh cleanup and alignment, to facilitate **CAD recreation**, supporting 15+ customer projects with accurate digital models.
- Advised 30+ customers on integrating **3D printing and scanning** into their existing manufacturing workflows, providing technical input for pre-sales consultations.
- Documented **application notes and best practices** for professional and desktop 3D printers, improving onboarding efficiency for new team members by **20%**.
- Responded to **50+ customer support calls**, troubleshooting issues with 3D printers and software, and resolving 90% of reported problems on first contact.

SKILLS

3D Printing & Scanning – Additive Manufacturing Technologies, Prototype Part Production, Post-Processing, Mesh Cleanup, CAD Recreation, Reverse Engineering, 3D Scanning Demonstrations, Desktop 3D Printers, Professional 3D Printers

Software & Tools – SolidWorks, Fusion 360, Inventor, 3D File Preparation Software, Data Modelling, Power BI, Tableau, R, Python (pandas, NumPy, scikit-learn)

Engineering & Manufacturing – Materials Science, Product Design, Manufacturing Processes, Tolerances, Digital Workflows, Print File Optimization, Application Engineering

Technical Support & Operations – Troubleshooting, Customer Support, Lab Environment Maintenance, Process Documentation, Change Management, Production Support, SLA Management

Cloud & Automation – AWS (EC2, S3, Lambda, CloudWatch), Python Scripting, Jenkins, Git, Docker, Linux Shell Scripting

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of Ireland, Dublin **02/2026**

BACHELOR OF ENGINEERING in Computer Science – SNS College of Technology, Coimbatore, India **01/2021**

PROJECTS

Additive Manufacturing Workflow Integration Platform

- Developed a Python-based platform to automate the preparation and optimization of **3D print files** for diverse customer jobs, reducing manual processing time by **40%**.

- Integrated 3D scanning data ingestion and basic **reverse engineering** routines, enabling rapid conversion of physical parts into digital CAD models with **95% geometric accuracy**.
- Designed and implemented a customer-facing dashboard using **Power BI** to track job progress, manage print queues, and visualize key performance indicators for prototype production, improving client communication by **25%**.

Financial Services Transaction Monitoring Dashboard

- Engineered backend services in **Java Spring Boot** to process and monitor high-volume financial transactions, ensuring **99.9% uptime** for critical operational alerting.
- Developed **Python scripts** for automated data validation and reconciliation workflows across PostgreSQL and Oracle databases, identifying and resolving an average of 15 discrepancies per day.
- Managed incident response and production support for a **payments operations dashboard**, adhering to strict SLAs and contributing to audit trail maintenance in a GDPR-aware environment.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)